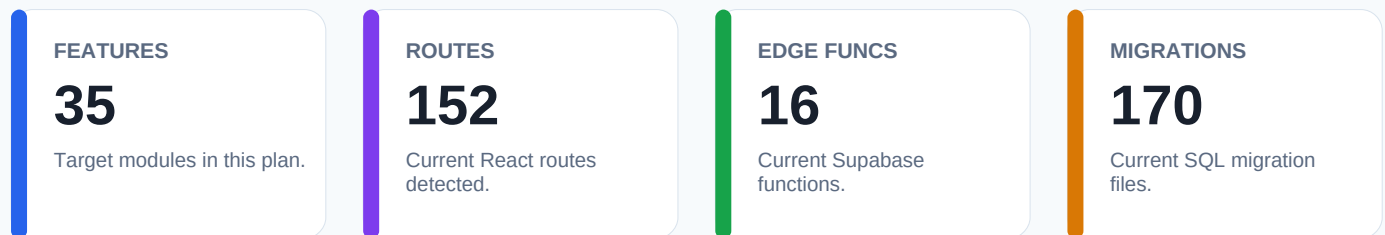


# XO Platform

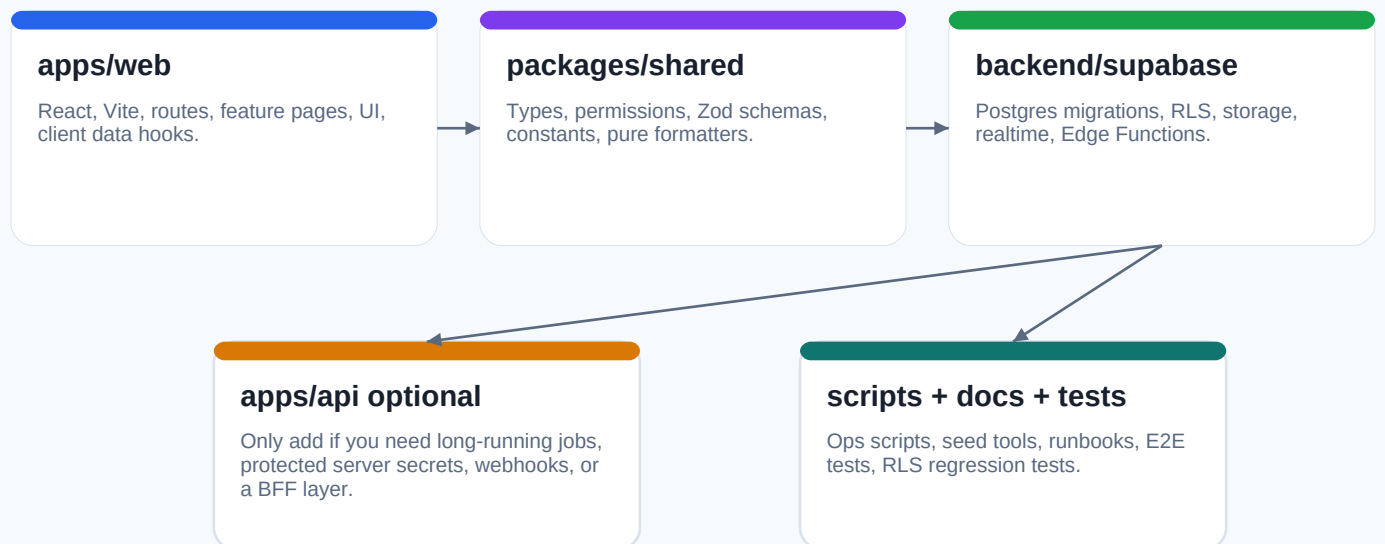
## Frontend / Backend Split - Graphic Project Structure

Generated from local repository on 2026-07-19



### Recommended Split

Use a monorepo split. The frontend lives in `apps/web`, Supabase lives in `backend/supabase`, shared contracts live in `packages/shared` and `packages/database`, and `apps/api` stays optional for server-only workflows that outgrow Edge Functions.



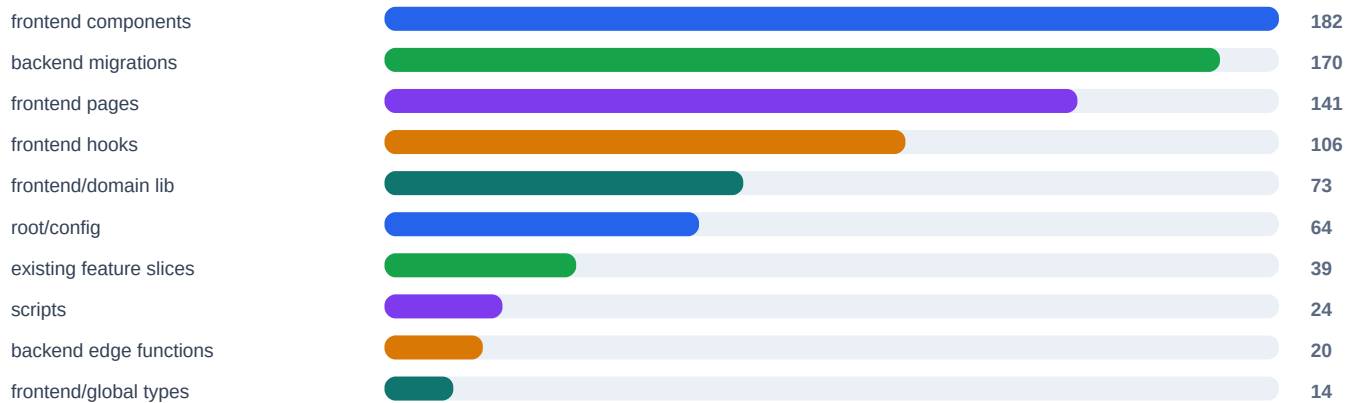
# Full Project File Structure

Target tree for splitting frontend and backend while keeping the project working as one repository.

```
xO-platform/  
  apps/  
    web/  
      public/  
      src/  
        app/  
          router/  
          providers/  
          layouts/  
        features/  
          auth-access/  
          platform-admin/  
          tenant-settings-rbac/  
          attendance/  
          events-permissions/  
          invoices-payments/  
          child-enrollment/  
          staff-onboarding/  
          ... all product features  
        shared/  
          ui/  
          components/  
          hooks/  
          lib/  
          types/  
        assets/  
        locales/  
        main.tsx  
        index.css  
      index.html  
      vite.config.ts  
      package.json  
    api/  
      src/  
        modules/  
        middleware/  
        config/  
        jobs/  
        tests/  
      package.json  
  backend/  
    supabase/  
      migrations/  
      functions/  
      seed/  
      config.toml  
  packages/  
    shared/  
      src/types/  
      src/constants/  
      src/permissions/  
      src/validation/  
      src/formatters/  
    database/  
      src/generated.types.ts  
      src/tables.ts  
      src/queryKeys.ts  
    config/  
      eslint/  
      typescript/  
      tailwind/  
  scripts/  
    dev/  
    seed/  
    ops/  
    reports/  
  docs/  
  architecture/  
  runbooks/
```

# Current to Target Mapping

How the current folders move into the split structure.



| Current location                      | Target location                              | Reason                                                         |
|---------------------------------------|----------------------------------------------|----------------------------------------------------------------|
| src/pages/*                           | apps/web/src/features/<feature>/pages/*      | Pages belong to product features, not only role folders.       |
| src/components/ui/*                   | apps/web/src/shared/ui/*                     | Design-system primitives stay shared in frontend.              |
| src/components/layouts/*              | apps/web/src/app/layouts/*                   | Layouts are app shell, not feature UI.                         |
| src/components/admin parent teacher/* | apps/web/src/features/<feature>/components/* | Move domain components beside the feature they support.        |
| src/hooks/useFeatureThing.ts          | apps/web/src/features/<feature>/hooks/*      | Feature hooks should live with their feature.                  |
| src/hooks/generic hooks               | apps/web/src/shared/hooks/*                  | Only app-agnostic hooks stay shared.                           |
| src/lib/domain helpers                | apps/web/src/features/<feature>/api or lib/* | Business logic belongs to the feature.                         |
| src/lib/permissions/*                 | packages/shared/src/permissions/*            | Permission rules should be shared across frontend and backend. |
| src/types/database.ts                 | packages/database/src/generated.types.ts     | Generated Supabase types should be a shared package.           |
| supabase/*                            | backend/supabase/*                           | Keep database, RLS, storage, and Edge Functions as backend.    |
| scripts/ops/*                         | scripts/ops/* or backend/scripts/*           | Ops remains outside frontend app.                              |

# Frontend App Structure

The web app should be a thin shell plus feature slices. Pages compose components and call hooks; hooks call feature API modules; shared contains only reusable primitives.

```
apps/web/src/  
  app/  
    router/  
      AppRouter.tsx  
      routes.admin.tsx  
      routes.parent.tsx  
      routes.teacher.tsx  
      routes.xo-admin.tsx  
    providers/  
      AppProviders.tsx  
      AuthProvider.tsx  
      DatabaseConnectionGate.tsx  
    layouts/  
      RootLayout.tsx  
      AdminLayout.tsx  
      ParentLayout.tsx  
      TeacherLayout.tsx  
      StaffLayout.tsx  
      XoAdminLayout.tsx  
  features/<feature>/  
    pages/  
    components/  
    hooks/  
    api/  
    schemas/  
    types.ts  
    index.ts  
    __tests__/  
  shared/  
    ui/  
    components/  
    hooks/  
    lib/  
    types/  
  locales/  
  assets/
```

| Frontend rule                        | Clean-code meaning                                                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| Pages stay thin                      | No large Supabase query chains or business decisions in page files.                        |
| Hooks own data state                 | React Query keys, loading states, mutations, and cache invalidation stay in feature hooks. |
| API modules own backend calls        | Supabase and Edge Function calls are typed and isolated in feature/api.                    |
| Components are local first           | A component is shared only after two or more features need it.                             |
| Router imports feature route modules | Avoid one giant router file that imports every screen directly.                            |

# Backend Structure

Supabase is already the backend. Move it under backend/supabase. Add apps/api only for workflows that need server runtime beyond Supabase Edge Functions.

```
backend/supabase/  
  migrations/  
    001_foundation.sql  
    YYYYMMDDHHMM_feature_change.sql  
  functions/  
    _shared/  
      admin.ts  
      dispatch.ts  
      http.ts  
    ai-assistant/  
    ai-import-mapper/  
    child-enrollment-complete/  
    email-dispatch/  
    emergency-broadcast/  
    event-qr-issue/  
    parent-signup-complete/  
    payment-reminders/  
    permission-deadline-reminder/  
    process-import/  
    qr-token/  
    qr-verify/  
    sms-dispatch/  
    staff-onboarding-complete/  
    tenant-export/  
    whatsapp-dispatch/  
  seed/  
  config.toml  
  
apps/api/                                # optional  
  src/modules/<domain>/  
    controller.ts  
    service.ts  
    repository.ts  
    validation.ts  
    routes.ts  
    types.ts
```

| Backend rule                             | Clean-code meaning                                                                                    |
|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Supabase is source of truth              | Migrations, RLS, storage policies, and Edge Functions belong under backend/supabase.                  |
| Use Edge Functions for event-driven work | Notifications, QR verification, imports, AI assistant, and tenant export can stay as Edge Functions.  |
| Add apps/api only with a reason          | Use it for webhooks, long jobs, server-only secrets, complex orchestration, or external integrations. |
| Generated types are shared               | Regenerate DB types into packages/database and import them from frontend/backend code.                |
| RLS tests belong with backend            | Add tests/rls for tenant isolation and parent/teacher/admin access rules.                             |

# Shared Packages

Shared code is what both frontend and backend need. Keep it small and boring: types, constants, validation schemas, permissions, and formatters.

```
packages/  
  shared/  
    src/  
      types/  
      constants/  
      permissions/  
        matrix.ts  
        can.ts  
        types.ts  
      validation/  
        childEnrollment.schema.ts  
        staffOnboarding.schema.ts  
        event.schema.ts  
        payment.schema.ts  
      formatters/  
        money.ts  
        dates.ts  
        names.ts  
      index.ts  
  database/  
    src/  
      generated.types.ts  
      tables.ts  
      queryKeys.ts  
      index.ts
```

| Put in shared package                      | Do not put in shared package                          |
|--------------------------------------------|-------------------------------------------------------|
| Role/permission constants                  | React pages or UI that belongs to one feature.        |
| Zod schemas reused by frontend and backend | Supabase client calls with feature-specific behavior. |
| Generated database types                   | Stateful React hooks tied to one screen.              |
| Pure formatters and enums                  | Business processes that belong to one domain module.  |
| Cross-app query key builders               | Large helper files that mix many features.            |

# All Feature Split Map

Every major project feature gets a frontend slice, backend ownership, and shared contract location.

| Feature                          | Frontend slice                                 | Backend owner                                        | Shared contract                                |
|----------------------------------|------------------------------------------------|------------------------------------------------------|------------------------------------------------|
| Auth and Access                  | apps/web/src/features/auth-access              | backend/supabase/auth policies                       | packages/shared/src/auth/types.ts              |
| XO Platform Admin                | apps/web/src/features/platform-admin           | backend/supabase/migrations/nurseries*.sql           | packages/shared/src/platform/types.ts          |
| Tenant Settings and RBAC         | apps/web/src/features/tenant-settings-rbac     | backend/supabase/migrations/rbac*.sql                | packages/shared/src/permissions/matrix.ts      |
| Admin Dashboard                  | apps/web/src/features/admin-dashboard          | backend/supabase/views/dashboard*.sql                | packages/shared/src/dashboard/types.ts         |
| Nursery Setup Onboarding         | apps/web/src/features/nursery-onboarding       | backend/supabase/migrations/onboarding*.sql          | packages/shared/src/onboarding/schema.ts       |
| Children and Profiles            | apps/web/src/features/children-profiles        | backend/supabase/migrations/children*.sql            | packages/shared/src/children/types.ts          |
| Child Enrollment                 | apps/web/src/features/child-enrollment         | backend/supabase/functions/child-enrollment-complete | packages/shared/src/child-enrollment/schema.ts |
| Parent Signup                    | apps/web/src/features/parent-signup            | backend/supabase/functions/parent-signup-complete    | packages/shared/src/parent-signup/schema.ts    |
| Admissions and Applications      | apps/web/src/features/admissions-applications  | backend/supabase/migrations/admissions*.sql          | packages/shared/src/admissions/types.ts        |
| Bulk Import                      | apps/web/src/features/bulk-import              | backend/supabase/functions/process-import            | packages/shared/src/import/validation.ts       |
| Attendance                       | apps/web/src/features/attendance               | backend/supabase/migrations/attendance*.sql          | packages/shared/src/attendance/types.ts        |
| Classes and Class Staff          | apps/web/src/features/classes                  | backend/supabase/migrations/class_staff*.sql         | packages/shared/src/classes/types.ts           |
| Courses                          | apps/web/src/features/courses                  | backend/supabase/migrations/courses*.sql             | packages/shared/src/courses/schema.ts          |
| Events and Permissions           | apps/web/src/features/events-permissions       | backend/supabase/functions/event-qr-issue            | packages/shared/src/events/schema.ts           |
| Invoices and Payments            | apps/web/src/features/invoices-payments        | backend/supabase/functions/payment-reminders         | packages/shared/src/billing/types.ts           |
| Financial Reports                | apps/web/src/features/financial-reports        | backend/supabase/views/finance*.sql                  | packages/shared/src/finance/formatters.ts      |
| Staff and HR                     | apps/web/src/features/staff-hr                 | backend/supabase/migrations/staff_profile*.sql       | packages/shared/src/staff/types.ts             |
| Staff Onboarding                 | apps/web/src/features/staff-onboarding         | backend/supabase/functions/staff-onboarding-complete | packages/shared/src/staff-onboarding/schema.ts |
| Payroll and Payslips             | apps/web/src/features/payroll-payslips         | backend/supabase/migrations/staff_payroll*.sql       | packages/shared/src/payroll/types.ts           |
| Media Upload and Gallery         | apps/web/src/features/media-gallery            | backend/supabase/migrations/media*.sql               | packages/shared/src/media/types.ts             |
| Daily Reports                    | apps/web/src/features/daily-reports            | backend/supabase/migrations/daily_report*.sql        | packages/shared/src/daily-reports/schema.ts    |
| Milestones and Quarterly Reports | apps/web/src/features/milestones-reports       | backend/supabase/migrations/milestones*.sql          | packages/shared/src/milestones/types.ts        |
| Health and Safety                | apps/web/src/features/health-safety            | backend/supabase/migrations/child_health*.sql        | packages/shared/src/health/schema.ts           |
| Chat and Messages                | apps/web/src/features/chat-messages            | backend/supabase/migrations/chat*.sql                | packages/shared/src/chat/types.ts              |
| Broadcasts and Notifications     | apps/web/src/features/broadcast-notifications  | backend/supabase/functions/email-dispatch            | packages/shared/src/notifications/types.ts     |
| Surveys and Forms                | apps/web/src/features/surveys-forms            | backend/supabase/migrations/surveys*.sql             | packages/shared/src/surveys/schema.ts          |
| Inventory                        | apps/web/src/features/inventory                | backend/supabase/migrations/inventory*.sql           | packages/shared/src/inventory/types.ts         |
| Meals and Nutrition              | apps/web/src/features/meals-nutrition          | backend/supabase/migrations/meals*.sql               | packages/shared/src/meals/types.ts             |
| Loyalty, Rewards, Packages       | apps/web/src/features/loyalty-rewards-packages | backend/supabase/migrations/loyalty*.sql             | packages/shared/src/loyalty/types.ts           |
| QR, Pickup, Check-in             | apps/web/src/features/qr-pickup-checkin        | backend/supabase/functions/qr-token                  | packages/shared/src/qr/types.ts                |

| Feature                       | Frontend slice                           | Backend owner                                        | Shared contract                        |
|-------------------------------|------------------------------------------|------------------------------------------------------|----------------------------------------|
| Community and Content Library | apps/web/src/features/community-content  | backend/supabase/migrations/community_content*.sql   | packages/shared/src/content/types.ts   |
| Personal Reminders            | apps/web/src/features/personal-reminders | backend/supabase/migrations/personal_reminders*.sql  | packages/shared/src/reminders/types.ts |
| AI Assistant and Help         | apps/web/src/features/ai-help            | backend/supabase/functions/ai-assistant              | packages/shared/src/ai/types.ts        |
| PWA, I18n, Theme              | apps/web/src/features/pwa-i18n-theme     | public/sw.js stays with web app; no backend required | packages/shared/src/i18n/types.ts      |
| Ops, Backup, Export           | apps/web/src/features/ops-backup-export  | backend/supabase/functions/tenant-export             | packages/shared/src/ops/types.ts       |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 1-3.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                     |                                                                                               |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------|--------|---------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <div>Auth and Access</div> <div>src/features/auth-access</div> <table><tr><td>Frontend</td><td>Shared</td><td>Backend</td></tr><tr><td>apps/web/src/features/auth-access/pages/LoginPage.tsx<br/>hooks/useAuthSession.ts<br/>components/AuthShell.tsx</td><td>packages/shared/src/auth/types.ts<br/>packages/shared/src/auth/roles.ts</td><td>backend/supabase/auth policies<br/>backend/supabase/functions/* when auth callbacks are needed</td></tr></table>                      |                                                                                     |                                                                                               | Frontend | Shared | Backend | apps/web/src/features/auth-access/pages/LoginPage.tsx<br>hooks/useAuthSession.ts<br>components/AuthShell.tsx     | packages/shared/src/auth/types.ts<br>packages/shared/src/auth/roles.ts              | backend/supabase/auth policies<br>backend/supabase/functions/* when auth callbacks are needed |
| Frontend                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Shared                                                                              | Backend                                                                                       |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |
| apps/web/src/features/auth-access/pages/LoginPage.tsx<br>hooks/useAuthSession.ts<br>components/AuthShell.tsx                                                                                                                                                                                                                                                                                                                                                                        | packages/shared/src/auth/types.ts<br>packages/shared/src/auth/roles.ts              | backend/supabase/auth policies<br>backend/supabase/functions/* when auth callbacks are needed |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |
| <div>XO Platform Admin</div> <div>src/features/platform-admin</div> <table><tr><td>Frontend</td><td>Shared</td><td>Backend</td></tr><tr><td>apps/web/src/features/platform-admin/pages/*<br/>components/NurseryForm.tsx<br/>hooks/useNurseries.ts</td><td>packages/shared/src/platform/types.ts</td><td>backend/supabase/migrations/nurseries*.sql<br/>backend/supabase/functions/tenant-export</td></tr></table>                                                                   |                                                                                     |                                                                                               | Frontend | Shared | Backend | apps/web/src/features/platform-admin/pages/*<br>components/NurseryForm.tsx<br>hooks/useNurseries.ts              | packages/shared/src/platform/types.ts                                               | backend/supabase/migrations/nurseries*.sql<br>backend/supabase/functions/tenant-export        |
| Frontend                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Shared                                                                              | Backend                                                                                       |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |
| apps/web/src/features/platform-admin/pages/*<br>components/NurseryForm.tsx<br>hooks/useNurseries.ts                                                                                                                                                                                                                                                                                                                                                                                 | packages/shared/src/platform/types.ts                                               | backend/supabase/migrations/nurseries*.sql<br>backend/supabase/functions/tenant-export        |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |
| <div>Tenant Settings and RBAC</div> <div>src/features/tenant-settings-rbac</div> <table><tr><td>Frontend</td><td>Shared</td><td>Backend</td></tr><tr><td>apps/web/src/features/tenant-settings-rbac/pages/*<br/>components/PermissionMatrix.tsx<br/>hooks/usePermissions.ts</td><td>packages/shared/src/permissions/matrix.ts<br/>packages/shared/src/permissions/can.ts</td><td>backend/supabase/migrations/rbac*.sql<br/>backend/supabase/migrations/roles*.sql</td></tr></table> |                                                                                     |                                                                                               | Frontend | Shared | Backend | apps/web/src/features/tenant-settings-rbac/pages/*<br>components/PermissionMatrix.tsx<br>hooks/usePermissions.ts | packages/shared/src/permissions/matrix.ts<br>packages/shared/src/permissions/can.ts | backend/supabase/migrations/rbac*.sql<br>backend/supabase/migrations/roles*.sql               |
| Frontend                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Shared                                                                              | Backend                                                                                       |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |
| apps/web/src/features/tenant-settings-rbac/pages/*<br>components/PermissionMatrix.tsx<br>hooks/usePermissions.ts                                                                                                                                                                                                                                                                                                                                                                    | packages/shared/src/permissions/matrix.ts<br>packages/shared/src/permissions/can.ts | backend/supabase/migrations/rbac*.sql<br>backend/supabase/migrations/roles*.sql               |          |        |         |                                                                                                                  |                                                                                     |                                                                                               |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 4-6.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <div><div>Admin Dashboard</div><div>src/features/admin-dashboard</div><div><div>Frontend</div><div>apps/web/src/features/admin-dashboard/pages/Dashboard.tsx<br/>components/StatsCards.tsx<br/>hooks/useDashboardStats.ts</div></div><div><div>Shared</div><div>packages/shared/src/dashboard/types.ts</div></div><div><div>Backend</div><div>backend/supabase/views/dashboard*.sql<br/>backend/supabase/rpc/dashboard*.sql</div></div></div>  |  |  |
| <div><div>Nursery Setup Onboarding</div><div>src/features/nursery-onboarding</div><div><div>Frontend</div><div>apps/web/src/features/nursery-onboarding/pages/steps/*<br/>components/OnboardingFrame.tsx</div></div><div><div>Shared</div><div>packages/shared/src/onboarding/schema.ts</div></div><div><div>Backend</div><div>backend/supabase/migrations/onboarding*.sql</div></div></div>                                                   |  |  |
| <div><div>Children and Profiles</div><div>src/features/children-profiles</div><div><div>Frontend</div><div>apps/web/src/features/children-profiles/pages/*<br/>components/ChildSelector.tsx<br/>hooks/useChildren.ts</div></div><div><div>Shared</div><div>packages/shared/src/children/types.ts</div></div><div><div>Backend</div><div>backend/supabase/migrations/children*.sql<br/>backend/supabase/storage/child-avatars</div></div></div> |  |  |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 7-9.

|                                                                                                                                                                                                                                                                          |                                                                                       |                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                                                             |                                                                                       |                                                                                                                                                |
| <div><div>Child Enrollment</div><div>src/features/child-enrollment</div><div><div>Frontend</div><div>apps/web/src/features/child-enrollment/pages/*<br/>components/ChildEnrollmentWizard.tsx<br/>steps/*.tsx</div></div></div>                                           | <div><div>Shared</div><div>packages/shared/src/child-enrollment/schema.ts</div></div> | <div><div>Backend</div><div>backend/supabase/functions/child-enrollment-complete<br/>backend/supabase/storage/enrollment-documents</div></div> |
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                                                             |                                                                                       |                                                                                                                                                |
| <div><div>Parent Signup</div><div>src/features/parent-signup</div><div><div>Frontend</div><div>apps/web/src/features/parent-signup/pages/ParentSignUpPage.tsx<br/>components/signup/*</div></div></div>                                                                  | <div><div>Shared</div><div>packages/shared/src/parent-signup/schema.ts</div></div>    | <div><div>Backend</div><div>backend/supabase/functions/parent-signup-complete<br/>backend/supabase/storage/signup-documents</div></div>        |
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                                                             |                                                                                       |                                                                                                                                                |
| <div><div>Admissions and Applications</div><div>src/features/admissions-applications</div><div><div>Frontend</div><div>apps/web/src/features/admissions-applications/pages/*<br/>components/InquiryKanban.tsx<br/>components/ApplicationReviewTabs.tsx</div></div></div> | <div><div>Shared</div><div>packages/shared/src/admissions/types.ts</div></div>        | <div><div>Backend</div><div>backend/supabase/migrations/admissions*.sql<br/>backend/supabase/migrations/applications*.sql</div></div>          |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 10-12.

|                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
| <div><div><div>Bulk Import</div><div>src/features/bulk-import</div><div>Frontend</div><div>apps/web/src/features/bulk-import/pages/*<br/>components/ImportWizard.tsx<br/>components/ColumnMapper.tsx</div></div><div><div>Shared</div><div>packages/shared/src/import/validation.ts</div></div><div><div>Backend</div><div>backend/supabase/functions/process-import<br/>backend/supabase/functions/ai-import-mapper</div></div></div> |  |  |
| <div><div><div>Attendance</div><div>src/features/attendance</div><div>Frontend</div><div>apps/web/src/features/attendance/pages/*<br/>components/AttendanceDailyBars.tsx<br/>hooks/useAttendance.ts</div></div><div><div>Shared</div><div>packages/shared/src/attendance/types.ts</div></div><div><div>Backend</div><div>backend/supabase/migrations/attendance*.sql<br/>backend/supabase/rpc/attendance*.sql</div></div></div>        |  |  |
| <div><div><div>Classes and Class Staff</div><div>src/features/classes</div><div>Frontend</div><div>apps/web/src/features/classes/pages/*<br/>components/ClassStaffEditor.tsx<br/>hooks/useClassStaff.ts</div></div><div><div>Shared</div><div>packages/shared/src/classes/types.ts</div></div><div><div>Backend</div><div>backend/supabase/migrations/class_staff*.sql<br/>backend/supabase/rpc/set_class_lead.sql</div></div></div>   |  |  |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 13-15.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <div><div>Courses</div><div>src/features/courses</div><div><div>Frontend</div><div>apps/web/src/features/courses/pages/*<br/>components/CourseCard.tsx<br/>hooks/useCourses.ts</div></div><div><div>Shared</div><div>packages/shared/src/courses/schema.ts</div></div><div><div>Backend</div><div>backend/supabase/migrations/courses*.sql</div></div></div>                                                                                                                |  |  |
| <div><div>Events and Permissions</div><div>src/features/events-permissions</div><div><div>Frontend</div><div>apps/web/src/features/events-permissions/pages/*<br/>components/EventFormFields.tsx<br/>hooks/useEventPermissions.ts</div></div><div><div>Shared</div><div>packages/shared/src/events/schema.ts</div></div><div><div>Backend</div><div>backend/supabase/functions/event-qr-issue<br/>backend/supabase/functions/permission-deadline-reminder</div></div></div> |  |  |
| <div><div>Invoices and Payments</div><div>src/features/invoices-payments</div><div><div>Frontend</div><div>apps/web/src/features/invoices-payments/pages/*<br/>components/InvoiceCard.tsx<br/>hooks/useInvoices.ts</div></div><div><div>Shared</div><div>packages/shared/src/billing/types.ts</div></div><div><div>Backend</div><div>backend/supabase/functions/payment-reminders<br/>apps/api/src/modules/payments if Paymob secret flow grows</div></div></div>           |  |  |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 16-18.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Financial Reports</div><div>src/features/financial-reports</div><div><div>Frontend</div><div>apps/web/src/features/financial-reports/pages/*<br/>components/charts/*.tsx<br/>hooks/useFinancialReports.ts</div></div><div><div>Shared</div><div>packages/shared/src/finance/formatters.ts</div></div><div><div>Backend</div><div>backend/supabase/views/finance*.sql<br/>backend/supabase/rpc/finance*.sql</div></div></div>                       |
| <div><div>Staff and HR</div><div>src/features/staff-hr</div><div><div>Frontend</div><div>apps/web/src/features/staff-hr/pages/*<br/>components/StaffDirectoryTable.tsx<br/>hooks/useStaff.ts</div></div><div><div>Shared</div><div>packages/shared/src/staff/types.ts</div></div><div><div>Backend</div><div>backend/supabase/migrations/staff_profiles*.sql<br/>backend/supabase/migrations/manager_hr*.sql</div></div></div>                               |
| <div><div>Staff Onboarding</div><div>src/features/staff-onboarding</div><div><div>Frontend</div><div>apps/web/src/features/staff-onboarding/pages/*<br/>components/StaffOnboardingWizard.tsx<br/>steps/*.tsx</div></div><div><div>Shared</div><div>packages/shared/src/staff-onboarding/schema.ts</div></div><div><div>Backend</div><div>backend/supabase/functions/staff-onboarding-complete<br/>backend/supabase/storage/staff-documents</div></div></div> |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 19-21.

|                                                                                                                                                                                                                                    |                                                                                    |                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                       |                                                                                    |                                                                                                                   |
| <div><div>Payroll and Payslips</div><div>src/features/payroll-payslips</div><div><div>Frontend</div><div>apps/web/src/features/payroll-payslips/pages/*<br/>components/PayslipForm.tsx<br/>hooks/usePayroll.ts</div></div></div>   | <div><div>Shared</div><div>packages/shared/src/payroll/types.ts</div></div>        | <div><div>Backend</div><div>backend/supabase/migrations/staff_payroll*.sql</div></div>                            |
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                       |                                                                                    |                                                                                                                   |
| <div><div>Media Upload and Gallery</div><div>src/features/media-gallery</div><div><div>Frontend</div><div>apps/web/src/features/media-gallery/pages/*<br/>components/MediaApprovalGrid.tsx<br/>hooks/useMedia.ts</div></div></div> | <div><div>Shared</div><div>packages/shared/src/media/types.ts</div></div>          | <div><div>Backend</div><div>backend/supabase/migrations/media*.sql<br/>backend/supabase/storage/media</div></div> |
| <div><div></div><div></div><div></div></div>                                                                                                                                                                                       |                                                                                    |                                                                                                                   |
| <div><div>Daily Reports</div><div>src/features/daily-reports</div><div><div>Frontend</div><div>apps/web/src/features/daily-reports/pages/*<br/>components/DailyReportForm.tsx<br/>sections/*.tsx</div></div></div>                 | <div><div>Shared</div><div>packages/shared/src/daily-reports/schema.ts</div></div> | <div><div>Backend</div><div>backend/supabase/migrations/daily_reports*.sql</div></div>                            |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 22-24.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                          |                                                                                                                     |                                                                                                                                 |                                                          |                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <div><div></div><div></div><div></div></div> <h2>Milestones and Quarterly Reports</h2> <p>src/features/milestones-reports</p> <table><tr><td><b>Frontend</b><br/>apps/web/src/features/milestones-reports/pages/*<br/>components/MilestoneCard.tsx<br/>hooks/useMilestones.ts</td><td><b>Shared</b><br/>packages/shared/src/milestones/types.ts</td><td><b>Backend</b><br/>backend/supabase/migrations/milestones*.sql<br/>backend/supabase/migrations/quarterly_reports*.sql</td></tr></table> |                                                          |                                                                                                                     | <b>Frontend</b><br>apps/web/src/features/milestones-reports/pages/*<br>components/MilestoneCard.tsx<br>hooks/useMilestones.ts   | <b>Shared</b><br>packages/shared/src/milestones/types.ts | <b>Backend</b><br>backend/supabase/migrations/milestones*.sql<br>backend/supabase/migrations/quarterly_reports*.sql |
| <b>Frontend</b><br>apps/web/src/features/milestones-reports/pages/*<br>components/MilestoneCard.tsx<br>hooks/useMilestones.ts                                                                                                                                                                                                                                                                                                                                                                   | <b>Shared</b><br>packages/shared/src/milestones/types.ts | <b>Backend</b><br>backend/supabase/migrations/milestones*.sql<br>backend/supabase/migrations/quarterly_reports*.sql |                                                                                                                                 |                                                          |                                                                                                                     |
| <div><div></div><div></div><div></div></div> <h2>Health and Safety</h2> <p>src/features/health-safety</p> <table><tr><td><b>Frontend</b><br/>apps/web/src/features/health-safety/pages/*<br/>components/HealthOverviewCard.tsx<br/>components/dialogs/*.tsx</td><td><b>Shared</b><br/>packages/shared/src/health/schema.ts</td><td><b>Backend</b><br/>backend/supabase/migrations/child_health*.sql<br/>backend/supabase/storage/health-documents</td></tr></table>                             |                                                          |                                                                                                                     | <b>Frontend</b><br>apps/web/src/features/health-safety/pages/*<br>components/HealthOverviewCard.tsx<br>components/dialogs/*.tsx | <b>Shared</b><br>packages/shared/src/health/schema.ts    | <b>Backend</b><br>backend/supabase/migrations/child_health*.sql<br>backend/supabase/storage/health-documents        |
| <b>Frontend</b><br>apps/web/src/features/health-safety/pages/*<br>components/HealthOverviewCard.tsx<br>components/dialogs/*.tsx                                                                                                                                                                                                                                                                                                                                                                 | <b>Shared</b><br>packages/shared/src/health/schema.ts    | <b>Backend</b><br>backend/supabase/migrations/child_health*.sql<br>backend/supabase/storage/health-documents        |                                                                                                                                 |                                                          |                                                                                                                     |
| <div><div></div><div></div><div></div></div> <h2>Chat and Messages</h2> <p>src/features/chat-messages</p> <table><tr><td><b>Frontend</b><br/>apps/web/src/features/chat-messages/pages/*<br/>components/ChatPanel.tsx<br/>hooks/useChat.ts</td><td><b>Shared</b><br/>packages/shared/src/chat/types.ts</td><td><b>Backend</b><br/>backend/supabase/migrations/chat*.sql<br/>backend/supabase/realtime/chat publication</td></tr></table>                                                        |                                                          |                                                                                                                     | <b>Frontend</b><br>apps/web/src/features/chat-messages/pages/*<br>components/ChatPanel.tsx<br>hooks/useChat.ts                  | <b>Shared</b><br>packages/shared/src/chat/types.ts       | <b>Backend</b><br>backend/supabase/migrations/chat*.sql<br>backend/supabase/realtime/chat publication               |
| <b>Frontend</b><br>apps/web/src/features/chat-messages/pages/*<br>components/ChatPanel.tsx<br>hooks/useChat.ts                                                                                                                                                                                                                                                                                                                                                                                  | <b>Shared</b><br>packages/shared/src/chat/types.ts       | <b>Backend</b><br>backend/supabase/migrations/chat*.sql<br>backend/supabase/realtime/chat publication               |                                                                                                                                 |                                                          |                                                                                                                     |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 25-27.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                             |                                                                                                                         |                                                                                                                              |                                                             |                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <div><div></div><div></div><div></div></div> <h2>Broadcasts and Notifications</h2> <p>src/features/broadcast-notifications</p> <table><tr><td><b>Frontend</b><br/>apps/web/src/features/broadcast-notifications/pages/*<br/>components/NotificationCenterDrawer.tsx</td><td><b>Shared</b><br/>packages/shared/src/notifications/types.ts</td><td><b>Backend</b><br/>backend/supabase/functions/email-dispatch<br/>sms-dispatch<br/>whatsapp-dispatch<br/>emergency-broadcast</td></tr></table> |                                                             |                                                                                                                         | <b>Frontend</b><br>apps/web/src/features/broadcast-notifications/pages/*<br>components/NotificationCenterDrawer.tsx          | <b>Shared</b><br>packages/shared/src/notifications/types.ts | <b>Backend</b><br>backend/supabase/functions/email-dispatch<br>sms-dispatch<br>whatsapp-dispatch<br>emergency-broadcast |
| <b>Frontend</b><br>apps/web/src/features/broadcast-notifications/pages/*<br>components/NotificationCenterDrawer.tsx                                                                                                                                                                                                                                                                                                                                                                            | <b>Shared</b><br>packages/shared/src/notifications/types.ts | <b>Backend</b><br>backend/supabase/functions/email-dispatch<br>sms-dispatch<br>whatsapp-dispatch<br>emergency-broadcast |                                                                                                                              |                                                             |                                                                                                                         |
| <div><div></div><div></div><div></div></div> <h2>Surveys and Forms</h2> <p>src/features/surveys-forms</p> <table><tr><td><b>Frontend</b><br/>apps/web/src/features/surveys-forms/pages/*<br/>components/FormBuilder.tsx<br/>components/FormRenderrer.tsx</td><td><b>Shared</b><br/>packages/shared/src/surveys/schema.ts</td><td><b>Backend</b><br/>backend/supabase/migrations/surveys*.sql</td></tr></table>                                                                                 |                                                             |                                                                                                                         | <b>Frontend</b><br>apps/web/src/features/surveys-forms/pages/*<br>components/FormBuilder.tsx<br>components/FormRenderrer.tsx | <b>Shared</b><br>packages/shared/src/surveys/schema.ts      | <b>Backend</b><br>backend/supabase/migrations/surveys*.sql                                                              |
| <b>Frontend</b><br>apps/web/src/features/surveys-forms/pages/*<br>components/FormBuilder.tsx<br>components/FormRenderrer.tsx                                                                                                                                                                                                                                                                                                                                                                   | <b>Shared</b><br>packages/shared/src/surveys/schema.ts      | <b>Backend</b><br>backend/supabase/migrations/surveys*.sql                                                              |                                                                                                                              |                                                             |                                                                                                                         |
| <div><div></div><div></div><div></div></div> <h2>Inventory</h2> <p>src/features/inventory</p> <table><tr><td><b>Frontend</b><br/>apps/web/src/features/inventory/pages/InventoryPage.tsx<br/>hooks/useInventory.ts</td><td><b>Shared</b><br/>packages/shared/src/inventory/types.ts</td><td><b>Backend</b><br/>backend/supabase/migrations/inventory*.sql</td></tr></table>                                                                                                                    |                                                             |                                                                                                                         | <b>Frontend</b><br>apps/web/src/features/inventory/pages/InventoryPage.tsx<br>hooks/useInventory.ts                          | <b>Shared</b><br>packages/shared/src/inventory/types.ts     | <b>Backend</b><br>backend/supabase/migrations/inventory*.sql                                                            |
| <b>Frontend</b><br>apps/web/src/features/inventory/pages/InventoryPage.tsx<br>hooks/useInventory.ts                                                                                                                                                                                                                                                                                                                                                                                            | <b>Shared</b><br>packages/shared/src/inventory/types.ts     | <b>Backend</b><br>backend/supabase/migrations/inventory*.sql                                                            |                                                                                                                              |                                                             |                                                                                                                         |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 28-30.

|                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <div>Meals and Nutrition</div> <div>src/features/meals-nutrition</div> <div><div>Frontend</div><div>apps/web/src/features/meals-nutrition/pages/*<br/>components/MealsSection.tsx<br/>hooks/useMealPlans.ts</div></div> <div><div>Shared</div><div>packages/shared/src/meals/types.ts</div></div> <div><div>Backend</div><div>backend/supabase/migrations/meals*.sql</div></div>                                                      |  |  |
| <div>Loyalty, Rewards, Packages</div> <div>src/features/loyalty-rewards-packages</div> <div><div>Frontend</div><div>apps/web/src/features/loyalty-rewards-packages/pages/*<br/>components/LoyaltyDashboard.tsx</div></div> <div><div>Shared</div><div>packages/shared/src/loyalty/types.ts</div></div> <div><div>Backend</div><div>backend/supabase/migrations/loyalty*.sql<br/>backend/supabase/migrations/packages*.sql</div></div> |  |  |
| <div>QR, Pickup, Check-in</div> <div>src/features/qr-pickup-checkin</div> <div><div>Frontend</div><div>apps/web/src/features/qr-pickup-checkin/pages/*<br/>components/ChildQRCodeCard.tsx<br/>components/QRScanner.tsx</div></div> <div><div>Shared</div><div>packages/shared/src/qr/types.ts</div></div> <div><div>Backend</div><div>backend/supabase/functions/qr-token<br/>backend/supabase/functions/qr-verify</div></div>        |  |  |

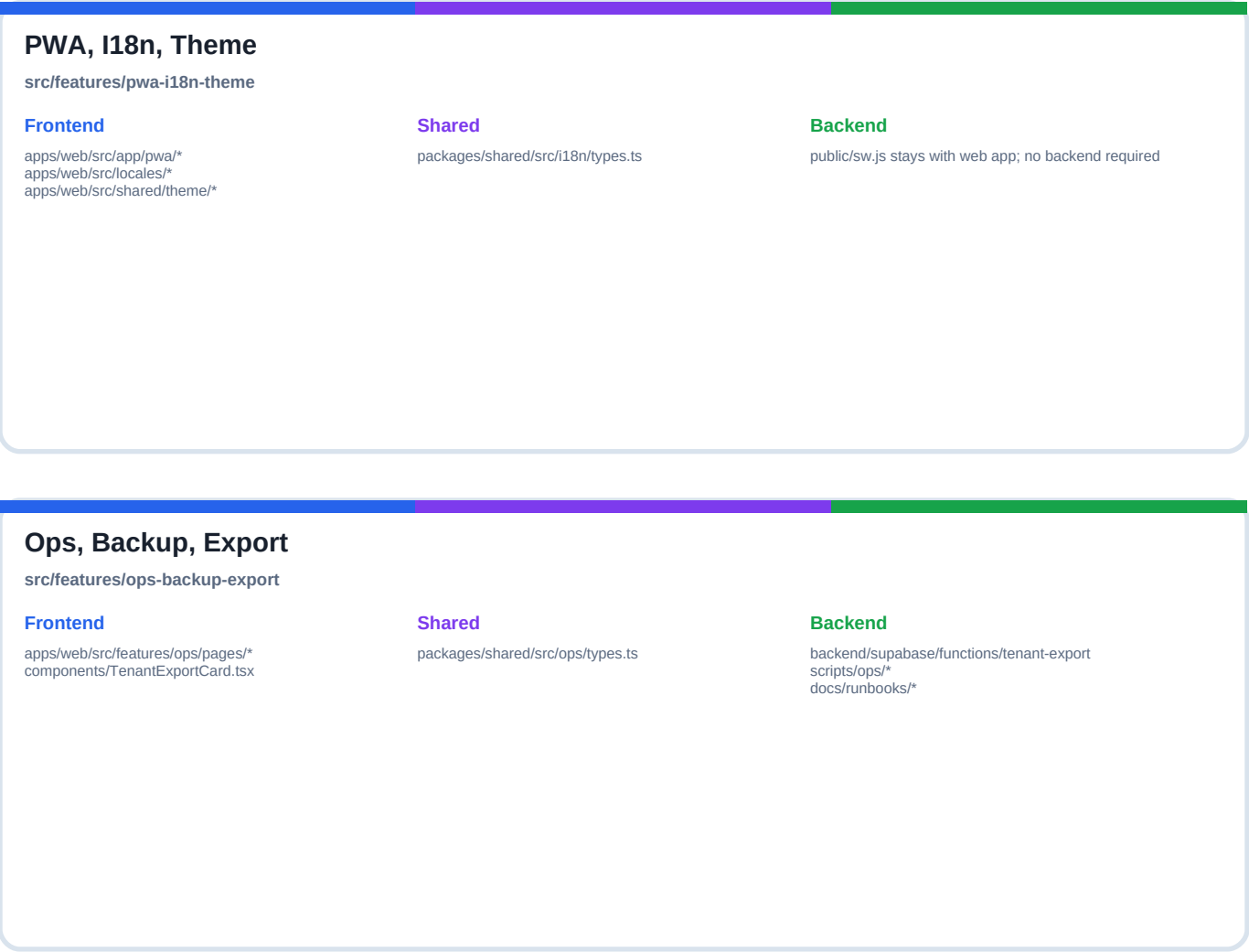
# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 31-33.

| <div>Community and Content Library</div> <div>src/features/community-content</div> <div><div>Frontend</div><div>apps/web/src/features/community-content/pages/*<br/>hooks/useCommunity.ts<br/>hooks/useContentLibrary.ts</div></div> <div><div>Shared</div><div>packages/shared/src/content/types.ts</div></div> <div><div>Backend</div><div>backend/supabase/migrations/community_content*.sql</div></div>                 |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <div>Personal Reminders</div> <div>src/features/personal-reminders</div> <div><div>Frontend</div><div>apps/web/src/features/personal-reminders/pages/RemindersPage.tsx<br/>components/ReminderAlertHost.tsx</div></div> <div><div>Shared</div><div>packages/shared/src/reminders/types.ts</div></div> <div><div>Backend</div><div>backend/supabase/migrations/personal_reminders*.sql</div></div>                           |  |  |
| <div>AI Assistant and Help</div> <div>src/features/ai-help</div> <div><div>Frontend</div><div>apps/web/src/features/ai-help/components/AIAssistant.tsx<br/>components/HelpPanel.tsx<br/>hooks/useAiAssistant.ts</div></div> <div><div>Shared</div><div>packages/shared/src/ai/types.ts<br/>packages/shared/src/help/articles.ts</div></div> <div><div>Backend</div><div>backend/supabase/functions/ai-assistant</div></div> |  |  |

# Feature-by-Feature Graphic Split

Frontend, shared, and backend structure for features 34-35.



# Migration Plan

Recommended order to split the project without stopping development.

| Wave | Move                                                                            | Done when                                                                 |
|------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1    | Create apps/web, backend/supabase, packages/shared, packages/database folders.  | Existing app still runs from the new apps/web location.                   |
| 2    | Move app shell: router, providers, layouts, main.tsx, index.css, public assets. | Routes render and auth/layouts still work.                                |
| 3    | Move Supabase backend into backend/supabase and update scripts.                 | Migrations/functions commands point to the new backend path.              |
| 4    | Generate DB types into packages/database.                                       | Frontend imports database types from one package and type drift drops.    |
| 5    | Move high-risk feature slices: events, payments, attendance, onboarding.        | Each feature has pages/components/hooks/api/schemas/tests together.       |
| 6    | Move remaining features in batches.                                             | No feature-specific logic remains in global hooks/lib/components folders. |
| 7    | Add apps/api only if needed.                                                    | There is a clear server-only use case, not just folder neatness.          |
| 8    | Add CI gates for typecheck, lint, route smoke, and RLS tests.                   | Structure stays clean after migration.                                    |

## Root package scripts after split

```
package.json
scripts:
  dev:web      -> npm --workspace apps/web run dev
  build:web    -> npm --workspace apps/web run build
  typecheck    -> npm run typecheck --workspaces
  lint         -> npm run lint --workspaces
  supabase:start -> supabase start --workdir backend/supabase
  db:types     -> generate packages/database/src/generated.types.ts
  test:e2e     -> playwright test tests/e2e
  test:rls     -> run SQL regression checks
```

# Appendix

Current repository facts used to build the graphic.

| Item                        | Detected value |
|-----------------------------|----------------|
| Package                     | xo-platform    |
| React                       | ^19.2.4        |
| Vite                        | ^8.0.1         |
| Supabase client             | ^2.100.0       |
| React Router                | ^7.13.2        |
| React Query                 | ^5.95.2        |
| Text/code files scanned     | 815            |
| Approximate text/code lines | 123,506        |
| Routes detected             | 152            |
| Supabase edge functions     | 16             |
| Supabase migrations         | 170            |

## Current Edge Functions

| Function folders             |
|------------------------------|
| ai-assistant                 |
| ai-import-mapper             |
| child-enrollment-complete    |
| email-dispatch               |
| emergency-broadcast          |
| event-qr-issue               |
| parent-signup-complete       |
| payment-reminders            |
| permission-deadline-reminder |
| process-import               |
| qr-token                     |
| qr-verify                    |
| sms-dispatch                 |
| staff-onboarding-complete    |
| tenant-export                |
| whatsapp-dispatch            |